United States Patent	12405775
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Guo; Peiran et al.

Mini program data binding method and apparatus, device, and storage medium

Abstract

This application discloses a mini program data binding method and apparatus, a device, and a non-transitory computer-readable storage medium. The mini program is a program executed in a host application program. The method includes: displaying a mini program production interface including a panel region and an editing region, the panel region being provided with n types of basic user interface (UI) controls, n being a positive integer; obtaining, when receiving a user operation on a selected basic UI control, a program interface of the mini program in the editing region according to the selected basic UI control; obtaining a data source; and when receiving a data binding operation corresponding to a target basic UI control on the program interface of the mini program, binding the target basic UI control with the data source according to the data binding operation, the data source being used for displaying the target basic UI control.

Inventors: **Guo; Peiran (Shenzhen, CN), Su; Haicheng (Shenzhen, CN), Cai; Yuli (Shenzhen, CN), Liu; Li (Shenzhen, CN), Zhu; Shida (Shenzhen, CN)**

Applicant: **Tencent Technology (Shenzhen) Company Limited** (Shenzhen, CN)

Family ID: **74111552**

Assignee: **TENCENT TECHNOLOGY (SHENZHEN) COMPANY LIMITED (Shenzhen, CN)**

Appl. No.: **18/386950**

Filed: **November 03, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20240061659 A1	Feb. 22, 2024

Foreign Application Priority Data

CN	201910636599.0	Jul. 15, 2019

Related U.S. Application Data

continuation parent-doc US 17361090 20210628 US 11853730 child-doc US 18386950
continuation parent-doc WO PCT/CN2020/098553 20200628 PENDING child-doc US 17361090

Publication Classification

Int. Cl.: **G06F8/38** (20180101); **G06F3/04842** (20220101); **G06F8/34** (20180101); **G06F9/448** (20180101); **G06F9/451** (20180101)

U.S. Cl.:

CPC **G06F8/38** (20130101); **G06F3/04842** (20130101); **G06F8/34** (20130101); **G06F9/448** (20180201); **G06F9/451** (20180201);

Field of Classification Search

CPC: G06F (8/38); G06F (9/448); G06F (9/451); G06F (3/04842); G06F (8/34)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7308674	12/2006	Fairweather	717/103	A61N 1/0553
8365138	12/2012	Iborra	717/113	G06F 16/2365
9176746	12/2014	Kothari	N/A	G06F 9/451
9329838	12/2015	Tattrie	N/A	G06F 8/34
9395958	12/2015	Slone	N/A	G06Q 10/06
11054971	12/2020	Zheng	N/A	G06F 3/04842
2001/0045963	12/2000	Marcos	715/765	G06F 9/451
2002/0065946	12/2001	Narayan	719/330	G06F 9/548
2003/0222912	12/2002	Fairweather	715/763	A61N 1/0553
2004/0036719	12/2003	Van Treeck	715/763	G06F 9/451
2005/0038796	12/2004	Carlson	707/E17.118	G06F 16/986
2005/0057560	12/2004	Bibr	345/418	G06F 9/451
2005/0066287	12/2004	Tattrie	715/769	G06F 8/34
2005/0114361	12/2004	Roberts	707/999.102	G06F 8/38
2005/0172261	12/2004	Yuknewicz	717/136	G06F 8/38
2005/0172264	12/2004	Yuknewicz	717/113	G06F 8/38
2005/0228801	12/2004	Peters	N/A	G06F 8/38
2005/0268280	12/2004	Fildebrandt	717/113	G06F 8/36
2006/0143592	12/2005	Bender	717/104	G06F 8/20
2006/0173863	12/2005	Paulus	N/A	H04L 67/02
2006/0212866	12/2005	Mckay	718/100	G06F 9/4843

2006/0235548	12/2005	Gaudette	700/83	G06F 8/34
2007/0030960	12/2006	Poole	380/28	H04L 9/0662
2007/0061740	12/2006	Marini	707/999.1	G06F 8/38
2007/0079282	12/2006	Nachnani	717/106	G06F 8/34
2007/0094604	12/2006	Sahoo	715/746	G06F 9/451
2007/0112714	12/2006	Fairweather	706/46	G06F 8/427
2007/0150385	12/2006	Ode	705/30	G06Q 40/12
2007/0156740	12/2006	Leland	707/999.102	G06F 16/2425
2007/0174482	12/2006	Yajima	709/238	G06F 40/174
2007/0217580	12/2006	Goose	379/88.06	G06F 3/016
2007/0256052	12/2006	Xu	717/110	G06F 8/38
2007/0282616	12/2006	Brunswig	705/348	G06Q 10/06
2008/0098289	12/2007	Williams	715/200	G06F 9/451
2008/0114810	12/2007	Malek	N/A	G06Q 10/10
2008/0270919	12/2007	Kulp	715/762	G06F 9/451
2009/0157627	12/2008	Arthursson	707/E17.127	H04L 67/02
2009/0172101	12/2008	Arthursson	709/205	G06F 3/0486
2009/0204820	12/2008	Brandenburg	713/182	G06Q 30/08
2009/0300047	12/2008	Carlson	707/999.102	G06F 16/00
2009/0300524	12/2008	Carlson	715/763	G06F 9/451
2009/0313601	12/2008	Baird	715/764	G06F 8/34
2009/0319921	12/2008	Abel	715/760	G06F 40/174
2011/0060704	12/2010	Rubin	706/12	G06N 5/04
2012/0029661	12/2011	Jones	700/17	G05B 19/41865
2012/0047130	12/2011	Perez	707/723	G06Q 10/00
2012/0059842	12/2011	Hille-Doering	707/769	G06F 16/3322
2012/0059872	12/2011	Chen	709/203	H04L 51/04
2012/0123996	12/2011	Krinsky	707/602	G06F 16/9577
2012/0290940	12/2011	Quine	715/744	G06F 8/34
2012/0290955	12/2011	Quine	715/763	G06F 8/34
2012/0290959	12/2011	Quine	715/765	G06F 9/451
2012/0291005	12/2011	Quine	717/105	G06F 9/45529
2012/0291006	12/2011	Quine	717/105	G06F 9/45529
2012/0291011	12/2011	Quine	717/115	G06F 9/45512
2013/0167050	12/2012	Colletti	715/762	G06F 8/38
2014/0033053	12/2013	Coloma Baiges	715/736	G06F 3/0484
2014/0289700	12/2013	Srinivasaraghavan	717/106	G06F 8/34
2014/0310273	12/2013	Mital	707/736	G06F 40/18
2015/0019991	12/2014	Kristj	715/747	H04L 41/0853
2015/0089469	12/2014	Shakespeare	717/106	G06F 9/44521

2015/0100946	12/2014	Brunswig	717/124	G06F 11/3664
2015/0169428	12/2014	Isman	717/124	G06F 11/36
2015/0227299	12/2014	Pourshahid	715/771	G06F 3/0486
2015/0248202	12/2014	Shankar	715/808	G06F 3/0484
2016/0103592	12/2015	Prophete	715/771	G06F 3/04845
2016/0378323	12/2015	Kresl	715/763	G06F 3/0482
2016/0378439	12/2015	Straub et al.	N/A	N/A
2017/0004638	12/2016	Csenteri	N/A	G06T 11/206
2017/0315786	12/2016	Rogers et al.	N/A	N/A
2018/0059921	12/2017	De Ligt	N/A	G05B 19/409
2018/0203674	12/2017	Dayanandan	N/A	G06F 8/35
2018/0285328	12/2017	Pivato	N/A	G06F 16/958
2018/0334767	12/2017	Kim	N/A	H04N 23/00
2019/0042581	12/2018	Aylett	N/A	G06F 16/252
2019/0332374	12/2018	Harner	N/A	G06F 8/34
2020/0034159	12/2019	Narayanaswamy	N/A	G06F 9/451
2020/0183664	12/2019	Lee	N/A	G06F 8/38
2021/0294583	12/2020	Guo	N/A	G06F 8/34
2021/0342147	12/2020	Singh	N/A	G06F 9/451
2021/0349700	12/2020	Guo et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	**Application Date**	**Country**	**CPC**
106557314	12/2016	CN	N/A
108228253	12/2017	CN	N/A
109218416	12/2018	CN	N/A
109542543	12/2018	CN	N/A
2002163251	12/2001	JP	N/A
2005038404	12/2004	JP	N/A
2005301994	12/2004	JP	N/A
2006526192	12/2005	JP	N/A
2015005241	12/2014	JP	N/A

OTHER PUBLICATIONS

Gu Ma Pao Wen Le, “Others' WeChat Applet Development Took a Month to Code and Produce, But I Only Took Three Minutes”, Jun. 25, 2018, 12 pgs., Retrieved from the Internet: https://www.sohu.com/a/237654005_100151340. cited by applicant

Tencent Technology, European Office Action, EP 20841164.5, Oct. 27, 2023, 5 pgs. cited by applicant

Tencent Technology, Extended European Search Report, EP20841164.5, Jul. 29, 2022, 9 pgs. cited by applicant

Tencent Technology, ISRWO, PCT/CN2020/098553, Sep. 30, 2020, 7 pgs. cited by applicant

Tencent Technology, IPRP, PCT/CN2020/098553, Jan. 18, 2022, 5 pgs. cited by applicant

Tencent Technology, Japanese Office Action, JP Patent Application No. 2021-547516, Aug. 1, 2022, 8 pgs. cited by applicant

---

Primary Examiner: Stanley; Jeremy L

Attorney, Agent or Firm: Morgan, Lewis & Bockius LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of U.S. patent application Ser. No. 17/361,090, entitled "DATA BINDING METHOD, APPARATUS, AND DEVICE OF MINI PROGRAM, AND STORAGE MEDIUM" filed on Jun. 28, 2021, which is a continuation application of PCT Patent Application No. PCT/CN2020/098553, entitled "DATA BINDING METHOD, APPARATUS, AND DEVICE OF MINI PROGRAM, AND STORAGE MEDIUM" filed on Jun. 28, 2020, which claims priority to Chinese Patent Application No. 201910636599.0, entitled "MINI PROGRAM DATA BINDING METHOD AND APPARATUS, DEVICE, AND STORAGE MEDIUM" and filed on Jul. 15, 2019, all of which are incorporated herein by reference in their entirety.

FIELD OF THE TECHNOLOGY
(1) This application relates to the field of visualization programing, and in particular, to a mini program data binding method and apparatus, a device, and a storage medium.
BACKGROUND OF THE DISCLOSURE
(2) A mini program is a program executed in a host application program, and a user may experience services brought by different mini programs by adding various mini programs to an installed host application program.
(3) Various mini programs may be developed by companies, service agencies, or even personal users (for example, programmers) and submitted to an application provider to provide different services for users. The application provider provides a host application program as a program platform. The mini program implements various functions through a large quantity of data. A developer binds a mini program and corresponding data separately, and therefore, a user can see different content (such as text and images) on a program interface of the mini program, and implement a function such as a page jumping function by triggering a specific page.
(4) Based on the foregoing situations, when developing a mini program, the developer needs to use professional developer tools and write a large quantity of code to implement data binding. Therefore, the mini program developing requires the developer to have more professional program compilation ability and is difficult to be promoted to more ordinary personal users to use.
SUMMARY
(5) Embodiments of this application provide a mini program data binding method and apparatus, a device, and a storage medium, and therefore, a user may implement a function of editing or modifying displayed content in a target basic UI control by changing a data source bound with the target basic UI control, or unbinding a binding relationship between the target basic UI control and the data source, without writing code, thereby improving efficiency of a user for producing a mini program. The technical solutions are as follows:
(6) According to an aspect of this application, a mini program data binding method is performed at a terminal, the mini program being a program executed in a host application program, and the method including: displaying a mini program production interface of a visualization production program including a panel region and an editing region, the panel region being provided with n types of basic user interface (UI) controls, n being a positive integer; obtaining, when receiving a user operation on a selected basic UI control, a program interface of the mini program in the editing region according to the selected basic UI control; obtaining a data source; binding, when receiving a data binding operation corresponding to a target basic UI control on the program interface of the mini program, the target basic UI control with the data source according to the data binding operation, the data source being used for displaying the target basic UI control; and in response to

receiving a preview operation, transmitting a program package of the mini program to the host application program for running and/or previewing.

(7) According to another aspect of this application, a mini program data binding apparatus is provided, the mini program being a program executed in a host application program, and the apparatus including: a display module, configured to display a mini program production interface of a visualization production program including a panel region and an editing region, the panel region being provided with n types of basic UI controls, n being a positive integer; a receiving module, configured to obtain, when receiving a user operation on a selected basic UI control, a program interface of the mini program in the editing region according to the selected basic UI control; and an obtaining module, configured to obtain a data source, the receiving module being further configured to bind, when receiving a data binding operation corresponding to a target basic UI control on the program interface of the mini program, the target basic UI control with the data source according to the data binding operation, the data source being used for displaying the target basic UI control.

(8) According to another aspect of this application, a computer device is provided, including a processor and a memory, the memory storing at least one instruction, the at least one program being loaded and executed by the processor to implement the mini program data binding method according to the foregoing aspect.

(9) According to another aspect of this application, a non-transitory computer-readable storage medium is provided, storing at least one instruction, the at least one instruction being loaded and executed by a processor of a computer device to implement the mini program data binding method according to the foregoing aspect.

(10) According to another aspect of this application, a computer program product or a computer program is provided, including computer instructions, the computer instructions being stored in a computer-readable storage medium. A processor of a computer device reads the computer instructions from the computer-readable storage medium, and executes the computer instructions, so that the computer device performs the mini program data binding method provided in illustrative implementations of the foregoing aspect.

(11) The technical solutions provided in the embodiments of this application achieve at least the following beneficial effects:

(12) An operation is performed on a mini program production interface of a visualization production program, a target basic UI control of the mini program is bound with a data source, and the target basic UI control undergone binding can display content corresponding to the data source. Therefore, a user may implement a function of editing or modifying displayed content in the target basic UI control by changing the data source bound with the target basic UI control, or unbinding a binding relationship between the target basic UI control and the data source, without writing code, thereby improving efficiency of a user for producing a mini program.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a schematic diagram of a mini program production interface of a visualization production program according to an exemplary embodiment of this application.

(2) FIG. **2** is a block diagram of an implementation environment according to an exemplary embodiment of this application.

(3) FIG. **3** is a flowchart of a mini program data binding method according to an exemplary embodiment of this application.

(4) FIG. **4** is a schematic diagram of a mini program production interface of a visualization production program according to another exemplary embodiment of this application.

(5) FIG. **5** is a schematic interface diagram of binding a control component with a data field according to an exemplary embodiment of this application.
(6) FIG. **6** is a flowchart of a method for binding a control component with a data field according to an exemplary embodiment of this application.
(7) FIG. **7** is a schematic interface diagram in which a control component is bound with a data field according to an exemplary embodiment of this application.
(8) FIG. **8** is a flowchart of a method for binding a control component with a data field in a random manner according to an exemplary embodiment of this application.
(9) FIG. **9** is a schematic interface diagram in which a control component is bound with a data field according to another exemplary embodiment of this application.
(10) FIG. **10** is a flowchart of a data binding method of a visualization production program according to another exemplary embodiment of this application.
(11) FIG. **11** is a schematic interface diagram of manually binding a control component with a data field according to an exemplary embodiment of this application.
(12) FIG. **12** is a schematic diagram of a mini program production interface in which a control component is bound with a data field according to an exemplary embodiment of this application.
(13) FIG. **13** is a schematic diagram of a preview interface of a mini program according to an exemplary embodiment of this application.
(14) FIG. **14** is a schematic diagram of a preview interface of a mini program on a terminal according to an exemplary embodiment of this application.
(15) FIG. **15** is a schematic interface diagram of performing data binding on a next hierarchy of program interface according to an exemplary embodiment of this application.
(16) FIG. **16** is a schematic interface diagram of performing data binding on a next hierarchy of program interface according to another exemplary embodiment of this application.
(17) FIG. **17** is a schematic diagram of a preview interface of a mini program according to another exemplary embodiment of this application.
(18) FIG. **18** is a schematic diagram of a preview interface of a mini program on a terminal according to another exemplary embodiment of this application.
(19) FIG. **19** is a schematic interface diagram of logic editing according to an exemplary embodiment of this application.
(20) FIG. **20** is a structural block diagram of a terminal according to an exemplary embodiment of this application.
(21) FIG. **21** is a schematic structural diagram of a mini program data binding apparatus according to an exemplary embodiment of this application.
(22) FIG. **22** is a schematic structural diagram of a computer device according to an exemplary embodiment of this application.

DESCRIPTION OF EMBODIMENTS

(23) First, terms involved in embodiments of this application are introduced:
(24) Visualization production program is a program that shows a program production process in a visualization preview manner to assist production. In some embodiments, the visualization production program may be a program programing through code and generating preview content in an editing region according to a code input, or may be a program automatically generating code data by directly adding interface content to the editing region and then editing the interface content. In this embodiment of this application, an example in which the visualization production program is a program automatically generating code data by adding interface content in the editing region and editing the interface content is used for description.
(25) In some embodiments, the visualization production program may be a program producing a mini program, or may be a program producing an ordinary program. In this embodiment of this application, an example in which the visualization production program produces a mini program is used for description.

(26) Basic user interface (UI) controls are any visible controls or elements that can be seen on a UI of an application, for example, controls such as an image, an input box, a text box, a button, and a label. Some UI controls respond to operations of a user. For example, the user can enter text in the input box, and the user performs information interaction with the UI by using the foregoing UI controls.
(27) Data is content on a page, for example, images, text, videos, audios, animations are all data. The data may be static or dynamic. The static data is data written into a program during programing, and the dynamic data is data obtained through a network protocol in a running process of the program. The network protocol refers to a set of rules, standards or agreements established for data exchange in a computer network. In the network, characters used by terminals are different, and therefore, a command entered by a user at a terminal cannot be recognized at another terminal. For an ordinary communication, the network protocol stipulates that each terminal needs to convert respective characters into standard characters before inputting the characters to a network for transmission, and then convert, after the standard characters reach a target terminal, the standard characters into characters used by the target terminal. In some embodiments, the network protocol is a transmission control protocol/internet protocol (TCP/IP).
(28) View layer is a development document of a mini program. When a mini program is run on a host application program, a development document of the mini program is rendered into a page on a screen, and static data may be directly written into the development document of the mini program.
(29) Logic layer is a file (a JavaScript file, also referred to as a JS file) written in an interpreted scripting language, is applicable to a mini program, and includes a dynamic logic of the mini program, for example, responding to a user click event and transmitting a request to obtain data. The logic layer modifies views through data binding.
(30) Data binding is to bind data with a corresponding UI control to generate an interface of a mini program. Dynamic data of the view layer needs to be provided by the logic layer. When writing a program corresponding to the view layer, the view layer and the logic layer are ensured to be in one-to-one correspondence with each other by reserving a position for the dynamic data.
(31) Data field means that each piece of data may include a plurality of data fields. For example, a piece of student information data includes “student number”, “name”, “gender”, “age”, “score”, and other data fields.
(32) Data element format is used for describing a format and type of data. For example, a class information table may be presented by using a list formed by data forming a “class”, each item of the list includes three data fields, namely, “name”, “age”, and “gender”.
(33) For example, an example in which the visualization production program produces a mini program is used for description in this application. The method provided in this application may be applied to the following scenario: An ordinary user with no program compilation knowledge binds basic UI controls with data sources by using a panel region and an editing region on a mini program production interface, to further complete the production of the mini program. Alternatively, a professional developer with program compilation knowledge binds basic UI controls with data sources in a visualization manner by using a mini program production interface and does not need to write a large quantity of code, thereby improving the efficiency of the professional developer to develop a mini program. In addition, code can also be generated by using the mini program interface of the visualization production program, thereby facilitating check by the professional from a perspective of code.
(34) The method provided in this application may be applied to a visualization production program or a program having a visualization production program function. An example in which the method is applied to a visualization production program is used in the following embodiment for description.
(35) FIG. **1** is a schematic diagram of an interface of binding data of a visualization production

program according to an exemplary embodiment of this application. A panel region **101**, an editing region **102**, an image control component **103**, a title control component **104**, a content control component **105**, an attribute option **106**, a data binding option **107**, a delete option **108**, and a more option **109** are displayed on a mini program production interface **10**.

(36) In some embodiments, the panel region **101** is provided with n types of basic UI controls, n being a positive integer. In some embodiments, the basic UI controls include at least one of the image control component **103**, text control components (such as the title control component **104** and the content control component **105**), a video control component, an audio control component, and an animation control component. A list or frame of the mini program is displayed in the editing region **102**, and data needs to be added to the list. In some embodiments, the data is at least one of images, text, videos, audios, and animation. That is, a binding operation is performed on a corresponding basic UI control and corresponding data, and therefore an interface after the mini program is produced is generated. In some embodiments, the attribute option **106** is used for viewing an attribute of a target basic UI control, or viewing an attribute of a data source bound with the target basic UI control. The attribute refers to parameters related to the target basic UI control or the data source bound with the target basic UI control, includes but not limited to: a name, quantity, size, shape (such as zooming, stretching, cutting), transparency, filter (parameters used for changing a hue of an image or a video frame), and sticker (referring to elements superimposed on an image or a video frame). The data binding option **107** is used for performing a binding operation on the target basic UI control and the data source. The delete option **108** is used for unbinding the bound target basic UI control and data source, or deleting a basic UI control from the mini program. The more option **109** is used for viewing other functions of the mini program production interface, or used for viewing other forms of a list corresponding to the mini program, or used for sharing the mini program interface. This is not limited in this application. A user binds a target basic UI control with a data source by using the data binding option **107**. For example, the user binds the image control component **103** with a corresponding image data source. The image control component **103** and the corresponding image data source may be bound by selecting the image control component **103**, obtaining a corresponding image data source, and selecting the data binding option **107**.

(37) FIG. **2** is a structural block diagram of a computer system according to an exemplary embodiment of this application. A computer system **100** includes a first terminal **120**, a server **140**, and a second terminal **160**.

(38) A visualization production program platform or a platform supporting a visualization production program is installed or run on the first terminal **120**. The first terminal **120** is a terminal used by a first user, and the first user uses the visualization production program platform on the first terminal **120** to produce a program or a mini program. For example, the first user uses the visualization production program platform on the first terminal **120** to produce a mini program by binding a target basic UI control with a corresponding data source.

(39) The first terminal **120** is connected to the server **140** via a wireless network or a wired network.

(40) The server **140** includes at least one of one server, a plurality of servers, a cloud computing platform, and a virtualization center. For example, the server **140** includes a processor **144** and a memory **142**. The memory **142** further includes a display module **1421**, a receiving module **1422**, and an obtaining module **1423**. The server **140** is configured to provide a backend service for the visualization production program, for example, provide a storage service and an updating service of a data source corresponding to a program or a mini program. In some embodiments, the server **140** is responsible for primary computing work, and the first terminal **120** and the second terminal **160** are responsible for secondary computing work; or the server **140** is responsible for secondary computing work, and the first terminal **120** and the second terminal **160** are responsible for primary computing work; or the server **140**, the first terminal **120**, and the second terminal **160** perform

collaborative computing by using a distributed computing architecture among each other.
(41) A host application program is installed and run on the second terminal **160**, and a mini program published by the first terminal **120** is added to the host application program, the second terminal **160** being a terminal used by a second user. In some embodiments, the second user logs into or registers an account on the mini program to implement an operation of accessing the mini program. In some embodiments, a visualization production program platform may be also installed or run on the second terminal **160**. A host application program may be also installed and run on the first terminal **120**, and a mini program published by the second terminal **160** or another terminal is added to the host application program. The first user logs into or registers an account on the mini program to implement an operation of accessing the mini program.
(42) In some embodiments, the visualization production program platforms installed on the first terminal **120** and the second terminal **160** are the same, or the visualization production program platforms installed on the two terminals are the same type of visualization production program platforms in different control systems. The first terminal **120** may be generally one of a plurality of terminals, and the second terminal **160** may be generally one of a plurality of terminals. In this embodiment, only the first terminal **120** and the second terminal **160** are used as an example for description. Device types of the first terminal **120** and the second terminal **160** are the same or different. The device type includes at least one of a smartphone, a tablet computer, an ebook reader, a moving picture experts group audio layer III (MP3) player, a moving picture experts group audio layer IV (MP4) player, a laptop computer, and a desktop computer. The following embodiment is described by using an example that the terminal includes a tablet computer and a smartphone.
(43) A person skilled in the art may learn that there may be more or fewer terminals. For example, there may be only one terminal, or there may be dozens of or hundreds of terminals or more. The quantity and the device type of the terminals are not limited in the embodiments of this application.
(44) FIG. **3** shows a mini program data binding method according to an exemplary embodiment of this application. The method is applied to the first terminal **120** or the second terminal **160** in the computer system shown in FIG. **2** or another terminal in the computer system. The method includes the following steps:
(45) Step **310**. Display a mini program production interface of a visualization production program, a panel region and an editing region being displayed on the mini program production interface, the panel region being provided with n types of basic UI controls.
(46) In some embodiments, a mini program production interface of the visualization production program is displayed on a terminal, and the terminal is a smartphone, a tablet computer, a notebook computer, or a desktop computer. For example, a mini program production interface of the visualization production program is displayed on a tablet computer, and a user performs an editing operation on the mini program by using an editing operation on a display screen of the tablet computer. In some embodiments, the editing operation is at least one of a tap operation, a double-tap operation, a drag operation, a slide operation, and a long press operation. This is not limited in this application.
(47) An example in which a panel region and an editing region are displayed on the mini program production interface is described with reference to FIG. **4**. FIG. **4** is a schematic diagram of a mini program production interface of a visualization production program according to an exemplary embodiment of this application. A panel region **101** and an editing region **102** are displayed on a mini program production interface **11**. In some embodiments, the panel region **101** is located at a left side of the mini program production interface **11**. In some embodiments, the panel region **101** is located at a right side, an upper side, or a lower side of the mini program production interface **11**. A position of the panel region **101** in the mini program production interface is not limited in this application.
(48) In some embodiments, the panel region **101** is provided with n types of basic UI controls, n being a positive integer. For example, if n is 3, there are three basic UI controls provided on the

panel region **101**. For example, the basic UI controls include: input boxes (including a single line input box and a multi-line input box), an option (such as a virtual button or a check box), time, a dynamic switch (supporting a jumping between pages), a page layout, and the like.
(49) In some embodiments, the user may name the produced mini program, name the mini program before producing the mini program, or name the mini program after the mini program is produced, or name the mini program at any time during producing the mini program. In some embodiments, a name of the mini program is located at an upper side of the mini program production interface, and the mini program is named by using an editing button (optionally, an editing button in a "pen" shape). In some embodiments, the name of the mini program is located at a right side of the mini program. This is not limited in this application.
(50) In some embodiments, a preview control **129** is displayed on the mini program production interface **11**. The user transmits, by selecting the preview control **129**, a program package of the mini program to a host application program platform for auditing or publishing.
(51) Step **320**. In response to receiving a user operation on a selected basic UI control, generate a program interface of the mini program in the editing region according to the selected basic UI control.
(52) As shown in FIG. **1**, the terminal obtains, in response to receiving a user operation on a selected basic UI control, a program interface of the mini program in the editing region **102** according to the selected basic UI control. In some embodiments, the selected basic UI control includes an image control component **103**, a title control component **104**, and a content control component **105**. In some embodiments, the selected basic UI control may be a video control component, an audio control component, or an animation control component. For example, the program interface of the mini program is obtained after typesetting or layout editing is performed on the foregoing selected basic UI control.
(53) In some embodiments, when producing a mini program on a mini program production interface of a visualization production program, the user may add a list item template by selecting a list item template provided by a visualization program platform, customizing a list item template, or calling another program (that can generate a list item template or provide a list item template) or a file including a list item template. The list item is a structure that can hold data in a mini program interface, for example, a list structure of an address book in a communication program. In some embodiments, the user may add a list item template by using a tap operation, a double-tap operation, a drag operation, a slide operation, or a long press operation.
(54) Step **330**. Obtain a data source.
(55) In some embodiments, the user may obtain the data source by using at least one of the following manners.
(56) Manner 1. Display a first table file on the mini program interface of the visualization production program and receive an addition operation of the user, the addition operation being used for adding the first table file to the visualization production program. In some embodiments, the addition operation is a tap operation, a double-tap operation, a drag operation, a slide operation, or a long press operation.
(57) For example, the data source is obtained by dragging the first table file. A file with a table file name being A is dragged to the mini program production interface, and the mini program production interface automatically recognizes a data source in the table file, and adds the data source to the mini program production interface, for example, adds a data source corresponding to an image to a data table or a data option.
(58) Manner 2. Receive a program calling operation, the program calling operation being used for calling a table application in the visualization production program, and the table application including a second table file.
(59) For example, the data source is obtained by calling a table application. For example, there is a calling option in the mini program production interface, and the table application (such as an Excel)

is opened by using the calling option. The mini program performs a data source recognition operation on a file B (a file name of the second table file is B) in the table application, and adds the data source to the data table or the data option in the mini program production interface.
(60) Manner 3. Receive a third table file by using a short range wireless communication technology.
(61) For example, the terminal receives the third table file by using the short range wireless communication technology. In some embodiments, the short range wireless communication technology is a Bluetooth technology, a Zigbee technology, an Airdrop technology, or the like. The received table file is opened at a platform of the visualization production program, and the data source in the table file is added to the data table or the data option in the mini program production interface. In some embodiments, the third table file may be alternatively transmitted to the visualization production program platform by another application, for example, a social application, a mail application, an instant messaging application, or the like.
(62) In some embodiments, the user may select one specific piece of data from the data table, or select all data from the data table. In some embodiments, the user may drag the data table to a corresponding position (that is, in the list item) in the mini program production interface by using a tap operation, a double-tap operation, a drag operation, a slide operation, or a long press operation.
(63) Step **340**. Bind, in response to receiving a data binding operation corresponding to a target basic UI control on the program interface of the mini program, the target basic UI control with the data source, the data source being used for displaying the target basic UI control.
(64) Step **340** is described with reference to FIG. **5**. A data table binding status **121**, a data table creation option **122**, a data table **123**, a save option **124**, and a cancel option **125**, and the image control component **103**, the title control component **104**, and the content control component **105** on the mini program interface are displayed on a data binding interface **12** (that is, the mini program production interface **12**).
(65) In some embodiments, the data table binding status **121** is a state that no data table is bound, or is a state that at least one data table is bound. In some embodiments, when at least one data table is bound, a name of the data table is displayed. In some embodiments, the data table includes at least one data field. For example, a name of the data table is text-8.xlsx. The data table includes nine data fields, respectively corresponding to nine list items on the mini program interface, and data information included in each data field is an image data field, a title data field, and a content data field.
(66) The data table creation option **122** is used for creating a data table on the mini program production interface **12**. In some embodiments, data in the created data table may be directly entered by a user through the mini program production interface **12**, or may be data obtained by adding a table file or calling a table application.
(67) The save option **124** is used for saving an operation of a user in the mini program production interface **12**, including at least one of an editing operation, a delete operation, and a modification operation. The cancel option **125** is used for exiting from the mini program production interface **12**, or canceling a data table binding operation.
(68) In conclusion, according to the method provided in this embodiment, a data source is bound with a UI control at a mini program production interface of a visualization production program, and therefore, a user may implement a process of producing a mini program without writing program code, and an ordinary user with no professional knowledge can also produce a mini program.
(69) FIG. **6** is a flowchart of a method for binding a data field with a control component according to an exemplary embodiment of this application. The method is applied to the first terminal **120** or the second terminal **160** in the computer system shown in FIG. **2** or another terminal in the computer system. The method includes the following steps:
(70) Step **601**. Display, in response to receiving a data binding operation, binding preview information that binds m data fields with m control components respectively according to a first

arrangement order.
(71) In some embodiments, a basic UI control has m corresponding control components, and a data source includes m data fields, m being a positive integer greater than 1. For example, if m is 3, the basic UI control has three corresponding control components, and the data source includes three data fields. In some embodiments, a type of the data field includes: an image type (ImageUrl) and a character string type (String).
(72) FIG. **7** is a schematic interface diagram of binding a control component with a data field according to an exemplary embodiment of this application. A data table binding status **121**, a data table **123**, a save option **124**, a cancel option **125**, and a field association manner **126**, and an image control component **103**, a title control component **104**, and a content control component **105** on a mini program interface are displayed on a binding interface **13**.
(73) For example, the data table binding status **121** is in a bound state, and a name of the data table is displayed on the mini program production interface. The data table **123** includes data that can be bound with a target basic UI control. In some embodiments, the data table **123** includes an image data field, a title data field, and a content data field, separately corresponding to the image control component and the text control component. In some embodiments, if the data table **123** includes a video data field, the video data field corresponds to a video control component. A user may view a dataset to view all data tables imported or added to the mini program production interface. For example, binding preview information is displayed on the mini program interface, the binding preview information being information binding three data fields with three control components according to a first arrangement order. A user checks whether binding between a data field and a control component is correct by viewing the binding preview information. For example, it is incorrect to bind an image data field with a text control component. In some embodiments, the first arrangement order is an image data field, a title data field, and a content data field.
(74) Step **602**. Display, in response to receiving a rearrangement operation, binding preview information that binds the m data fields with the m control components respectively according to an (i+1).sup.th arrangement order.
(75) In some embodiments, the field association manner **126** is a random manner or a manual manner. In some embodiments, the (i+1).sup.th arrangement order is an arrangement order different from an i.sup.th arrangement order, i being a positive integer with an initial value being 1, and the i.sup.th arrangement order being generated randomly or preset in advance.
(76) For example, the data table **123** has three data fields: an image data field, a title data field, and a content data field, and the three data fields respectively correspond to an image control component, a title control component, and a content control component. The three data fields are arranged and combined, and all arrangement combinations of the data fields are sorted according to an order. In some embodiments, the arrangement order is generated randomly, or is preset by a visualization production program platform in advance, or is preset by a user through a visualization production program platform in advance.
(77) In an example, an image data field is represented by A, a title data field is represented by B, and a content data field is represented by C. There are six arrangement combinations (ABC, ACB, BAC, BCA, CAB, and CBA) for the three data fields, and the six arrangement combinations are sorted in a random generation manner. A second arrangement order is different from a first arrangement order (an initial value of i is 1). For example, the first arrangement order is ABC, and the second arrangement order may be ACB, BAC, BCA, CAB, or CBA. Any arrangement order may be determined as the second arrangement order, as long as the arrangement order is different from the arrangement order of ABC.
(78) Step **603**. Bind, in response to receiving a binding confirmation operation, the m data fields with the m control components respectively.
(79) In some embodiments, the user binds the m data fields with the m control components respectively by triggering a binding button. Alternatively, the user respectively binds the m data

fields with the m control components directly after selecting the data table **123**. A manner of a binding confirmation operation of the user is not limited in this application.

(80) For example, data fields in the data table **123** are bound with control components in a first list item on the mini program interface, and the data table **123** undergone binding is displayed as a bound data table. Binding preview information is displayed on the first list item of the mini program interface, the binding preview information including an image (that is, an image, corresponding to the image data field in the data table **123**, displayed in the image control component **103**; for example, the image is a movie poster); a title or a topic (that is, content displayed in the title control component **104**; for example, the topic is a book, and a name of the book is *The Adventures of Xiao Zhang*); and a brief description or abstract of the topic content or the topic content (that is, content displayed in the content control component **105**, and the brief description of the topic content only shows a part of the topic content. For example, this is a book worthy of recommendation).

(81) Rearranging the m data fields is described with reference to FIG. **8**. FIG. **8** is a flowchart of a method for rearranging m data fields according to an exemplary embodiment of this application. The method is applied to the first terminal **120** or the second terminal **160** in the computer system shown in FIG. **2** or another terminal in the computer system. The method includes the following steps:

(82) Step **801**. Generate all arrangement orders of the m data fields.

(83) For example, if m is 3, there are six arrangement orders for the three data fields.

(84) Step **802**. Determine, in response to receiving a rearrangement operation, an arrangement order, different from an i.sup.th arrangement order, from all the arrangement orders randomly as an (i+1).sup.th arrangement order.

(85) For example, the three data fields and three control components are respectively bound in a random manner. In an example, the three data fields are respectively A, B, and C, and all the arrangement orders are ACB, ABC, BCA, BAC, CAB, and CBA. An arrangement order (i is 1) different from a first arrangement order is randomly selected from all the arrangement orders. For example, the first arrangement order is ACB, and BCA is selected as the second arrangement order, BCA being an arrangement order different from the arrangement order of ACB.

(86) Step **803**. Read a j.sup.th data field and a field type from the m data fields according to the (i+1).sup.th arrangement order, 1≤j≤m.

(87) For example, the (i+1).sup.th arrangement order is BCA, an arrangement order different from ACB (the i.sup.th arrangement order). A j.sup.th data field and a field type thereof are read from the three data fields according to the arrangement order of BCA. In some embodiments, 1≤j≤m, and j being a positive integer. For example, if j is 2, the second data field is read from BCA, that is, a data field represented by C and a corresponding field type. In an example, an image data field is represented by A, a title data field is represented by B, and a content data field is represented by C. A data field of C read from the arrangement order of BCA is the content data field, and the corresponding field type is a character string type.

(88) Step **804**. Determine a target control component satisfying a binding condition from the m control components.

(89) In some embodiments, the binding condition includes that: the control component is not yet bound with a data field; or the control component matches the field type of the j.sup.th data field; or the control component is not yet bound with a data field and matches the field type of the j.sup.th data field.

(90) For example, a target control component satisfying a binding condition is determined from the three control components. In an example, the binding condition includes that: the control component is not yet bound with a data field and matches a field type of the second data field (j is 2). In some embodiments, BCA is the (i+1).sup.th arrangement order, and the field type of the second field is a character string type.

(91) In an example, if the image control component **103** on the mini program interface does not match the field type of the second data field and a field type of the image control component **103** is an image field type, the image control component **103** is not the target control component. If the title control component **104** is bound with a title data field, the title control component **104** is not the target control component. If the content control component **105** is not yet bound with a data field and matches the field type of the second data field, the content control component **105** is the target control component.
(92) Step **805**. Bind the target control component with the j.sup.th data field.
(93) For example, after the content control component **105** is determined as the target control component, the target control component is bound with the second data field, that is, the content control component **105** is bound with the content data field.
(94) Step **806**. Display, after the m control components are all bound, the binding preview information that binds the m data fields with the m control components respectively.
(95) For example, the image control component **103** and the title control component **104** are respectively bound according to the foregoing step. In an example, in a case in which the second data field (the title data field) is bound with the content control component **105** and the title control component **104** is bound with the content data field, the image control component **103** is to be bound with the image data field. As shown in FIG. **9**, after the three control components are all bound, the binding preview information that binds the three data fields with the three control components respectively is displayed.
(96) FIG. **9** is a schematic interface diagram of binding with a target control component after m data fields are rearranged according to an exemplary embodiment of this application. A data table binding status **121**, a data table **123**, a save option **124**, a cancel option **125**, and a field association manner **126**, and an image control component **103**, a title control component **104**, and a content control component **105** on the mini program interface are displayed on a rebinding interface **14**.
(97) For example, the field association manner **126** selected by the user is a random manner, the binding preview information of the three data fields with the three control components is viewed, and when the user is not satisfied with the current binding preview information, the user reselects a field association, for example, taps a random manner button again. In some embodiments, the field association may alternatively change a current binding relationship between the data field and the control component by using a field association switching button. This is not limited in this application.
(98) In an example, a user taps a random manner button, and a terminal generates, in response to receiving a rearrangement operation on the mini program production interface, all arrangement orders of three data fields, selects and determines an arrangement order, different from a previous arrangement order (an i.sup.th arrangement order), from all the arrangement orders as a current arrangement order (an (i+1).sup.th arrangement order), the current arrangement order being an image data field, a title data field, and a content data field, and reads the second data field and a field type thereof (the second data field is a title data field and the field type is a character string type) from the three data fields according to the current arrangement order. Moreover, the content control component **105** is not yet bound with a data field and matches the field type of the second data field (is a field pertaining to a character string type), and therefore, the content control component **105** is determined as the target control component, and the content control component **105** is bound with the second data field (the title data field). In addition, the first data field and the third data field are respectively bound with corresponding control components: the first image data field is bound with the image control component **103**, and the third content data field is bound with the title control component **104**.
(99) In a mini program interface on the rebinding interface **14**, an image (that is, a cover image of a book) is displayed on the image control component **103**, a content abstract (for example, this is a book worthy of recommendation) is displayed on the title control component **104**, and a related

title (for example, *The Adventures of Xiao Zhang*) is displayed on the content control component **105**. Compared with the mini program interface shown in FIG. **7** on which the image control component **103**, the title control component **104**, and the content control component **105** are bound with the corresponding image data field, title data field, and content data field respectively, the title control component **104** and the content control component **105** shown in FIG. **9** are bound with the content data field and the title data field respectively.
(100) In conclusion, by sorting all arrangement combinations of data fields (an arrangement order is generated randomly or preset in advance), an (i+1).sup.th arrangement order is selected, the arrangement order being an arrangement order different from an i.sup.th arrangement order, so that binding relationships between the data fields and the control components are not repeated in a random manner for a user. It may be understood that, by randomly binding data fields with control components, the user can bind the data fields with the control components automatically according to the method provided in the foregoing embodiment only by triggering a rearrangement operation and without modifying a wrong binding between the data fields and the control components by using code.
(101) FIG. **10** is a flowchart of a random binding and manual binding method according to an exemplary embodiment of this application. The method is applied to the first terminal **120** or the second terminal **160** in the computer system shown in FIG. **2** or another terminal in the computer system. The method includes the following steps:
(102) Step **1001**: Start to bind.
(103) For example, a user performs a process of producing a mini program on a tablet computer visually, that is, edits on a mini program production interface on the tablet computer in a touch control manner.
(104) Step **1002**. Select a data table.
(105) The user selects a data table, the data table being a data source of bound data. In some embodiments, the data table includes at least one data field. In some embodiments, the user selects at least one data table according to content of the mini program. In some embodiments, the user performs a data table selection operation by adding a table file, adding data to a corresponding data table in the mini program production interface, or calling a table application on a terminal.
(106) For example, the user performs the data table selection operation by calling a table application on the tablet computer, the table application including at least one data field corresponding to the content of the mini program.
(107) Step **1003**. Generate all arrangement orders of m data fields.
(108) For example, if m is 3, the data table includes three data fields. After the data table selection operation of the user is received on the mini program production interface, all arrangement orders of the three data fields are generated, and there are six arrangement orders for the three data fields.
(109) Step **1004**. Disrupt the arrangement orders randomly.
(110) In some embodiments, all of the six arrangement orders of the three data fields are disrupted randomly. Alternatively, all of the six arrangement orders of the three data fields are arranged according to a preset arrangement order. In some embodiments, the preset arrangement order is an arrangement order preset (set by default) by a visualization production program platform in advance, or is an arrangement order set by a user according to personal habits or preferences by using a setting option autonomously, or is an arrangement order generated intelligently by a server according to a historical binding relationship record of the user.
(111) Step **1005**. Select a next arrangement order.
(112) In some embodiments, in a case that a rearrangement operation of the user is received on the mini program production interface, an arrangement order different from an i.sup.th arrangement order is randomly selected from all the arrangement orders and determined as an (i+1).sup.th arrangement order.
(113) For example, binding between the data field and the control component in the mini program

production interface is determined according to the first arrangement order in all the arrangement orders of the data fields. When the user selects a rearrangement operation (for example, taps a random option), an arrangement order different from the first arrangement order is randomly selected from all the arrangement orders. For example, the third arrangement order is selected from all the arrangement orders and determined as a next (i+1).sup.th arrangement order.
(114) Step **1006**. Match with a control component type.
(115) In some embodiments, the control component type includes at least one of a text type, an image type, a video type, an audio type, and an animation type. For example, the visualization production program platform matches an image data field with an image control component.
(116) Step **1007**. Establish a default binding relationship.
(117) The visualization production program platform binds a data field with a control component to establish a binding relationship. In some embodiments, the binding relationship is a default binding relationship of the visualization production program platform, and the default binding relationship is a default setting of the visualization production program platform to the binding relationship.
(118) Step **1008**. Switch a random sequence.
(119) For example, the user taps a random option to switch an arrangement order of the data fields, and step **1005** is performed; and otherwise, step **1009** is performed.
(120) Step **1009**. Adjust the binding relationship manually.
(121) In some embodiments, the user may further adjust a binding relationship between the data field and the control component manually. A process of adjusting the binding relationship manually is described with reference to FIG. **11**. FIG. **11** is a schematic interface diagram of a binding modification operation according to an exemplary embodiment of this application. A data table binding status **121**, a data table **123**, a save option **124**, a cancel option **125**, a field association manner **126**, and a candidate data field **127**, and an image control component **103**, a title control component **104**, and a content control component **105** on a mini program interface are displayed on a binding modification operating interface **15**.
(122) A binding modification operation on a target control component of a target basic UI control on the mini program production interface is received, and at least one candidate data field **127** corresponding to the target control component is displayed. In some embodiments, the candidate data field **127** may be displayed in a manner of a table or dialog box, and the candidate data field **127** is a data field, whose field type matches the target control component, in at most m data fields. In an example, the candidate data field **127** is displayed in a manner of a dialog box. A title option **1271**, a content option **1272**, and an unbinding option **1273** are displayed in a dialog box of the candidate data field **127**. The title option **1271** is used for matching a title data field with a title control component. The content option **1272** is used for matching a content data field with a content control component. The unbinding option **1273** is used for performing an unbinding operation on a binding relationship between a selected data field and control component.
(123) In some embodiments, the user taps the unbinding option **1273** to unbind a content data field and the content control component **105**, and then taps the content option **1272** to bind the content data field with the content control component **105**. Alternatively, the user directly taps the content option **1272** to bind the content data field with the content control component **105**. In some embodiments, binding the content data field with the content control component **105** may be binding a content data field which is imported to the visualization production program with the content control component **105**, or binding a new content data field reselected from the data table (that is, the data source) with the content control component **105**.
(124) Step **1010**. Complete binding between the data field and the control component.
(125) In a case that a selection operation is received, a data field selected from the at least one candidate data field is bound with the target control component. In some embodiments, the selection operation is a tap operation, a double-tap operation, a drag operation, a slide operation, or a long press operation.

(126) For example, the user taps the content option **1272** to bind the content data field with the content control component **105**.
(127) In some embodiments, after binding the data field with the control component, the user taps the save option **124** to save the produced mini program, exits from a data binding interface, and enters a mini program production interface **16**.
(128) FIG. **12** is a schematic interface diagram of saving a mini program according to an exemplary embodiment of this application. An image control component **103**, a title control component **104**, and a content control component **105**, a bound data field **128**, and a preview control **129** are displayed on the mini program production interface **16**.
(129) In some embodiments, the data field **128** in which data binding is completed is displayed on the mini program production interface **16**. In some embodiments, there are nine data fields, the data field is added by using a table file, and a name of the table file is text-8. The user taps the preview control **129**, the preview control **129** being used for pushing a program package of the mini program to a host application program for at least one operation of running and previewing. The previewing means that, when a mini program is produced but not published on the host application program, the user may browse or use the mini program in advance to check content or function of the mini program for errors. The running means to check whether the mini program can be normally used in the host application program.
(130) In some embodiments, the preview control is a triangle. In some embodiments, the preview control is a circle, a rectangle, a hexagon, or a polygon. Alternatively, the preview control is a virtual button displayed with "preview". A form of the preview control is not limited in this application.
(131) FIG. **13** is a schematic interface diagram of mini program preview prompt information according to an exemplary embodiment of this application. An image control component **103**, a title control component **104**, and a content control component **105**, a bound data field **128**, a preview control **129**, preview prompt information **130** are displayed on a mini program preview prompt interface **17**.
(132) In some embodiments, when the user taps the preview control **129**, the preview prompt information **130** is displayed; or when the user uses a terminal connected with an external device (for example, uses a desktop computer connected with a mouse), the preview prompt information **130** is displayed when an identifier corresponding to the external device is moved to the preview control; or when the user uses a function of previewing the mini program for the first time, the preview prompt information **130** is displayed for prompting the user. In an example, the user uses the visualization production program platform on the desktop computer, and when a mouse cursor is moved to the preview control **129**, the preview prompt information **130** is displayed. When the user clicks the preview control **129**, the produced mini program is previewed.
(133) In some embodiments, "The mini program is manually pushed to the host application program" is displayed on the preview prompt information **130**. In some embodiments, "pushed" is displayed on the preview prompt information **130**. This is not limited in this application.
(134) In some embodiments, the user may automatically push the mini program to the host application program for previewing by using manual pushing on the preview prompt information **130**. In some embodiments, the manual pushing may be that the user opens a preview interface of a mini program by copying a link of the mini program to a host application program, or may be that the user generates a graphic code corresponding to a preview interface of a mini program, and calls a camera in a terminal or a scanning function in a host application program to scan the graphic code to open the preview interface. This is not limited in this application.
(135) FIG. **14** is a schematic diagram of a preview interface of a mini program on a terminal according to an exemplary embodiment of this application. An image **201**, a title **202**, and content **203** are displayed on a mini program preview interface **200**. In an example, the image **201** is a cover of a book, the title **202** is a name of the book (for example, *The Adventures of Xiao Zhang*),

and the content **203** is a content brief description or a review of the book (for example, this is a book worthy of recommendation).
(136) In some embodiments, the mini program preview interface **200** further includes content displayed after other data fields and control components are bound. For example, there are six items of content displayed on the mini program preview interface **200**, and the six items of content being content generated after six data fields are bound with corresponding control components.
(137) In some embodiments, the mini program further includes a next hierarchy of program interface, the next hierarchy of program interface being a program interface displayed after a target basic UI control is triggered. The user produces the next hierarchy of program interface. FIG. **15** is a schematic diagram of a production interface of a next hierarchy of program interface according to an exemplary embodiment of this application. A completed mini program interface (that is, the mini program preview interface) **200**, a next hierarchy of program interface **220**, an attribute option **106**, a data binding option **107**, a delete option **108**, an operating option **111**, an image changing option **112**, and a preview control **129** are displayed on a production interface **18** of a next hierarchy of program interface.
(138) In some embodiments, the completed mini program interface **200** is the mini program interface shown in FIG. **14**. In some embodiments, the user taps the mini program interface **200** to generate a next hierarchy of program interface, or generate a next hierarchy of program interface by adding a list option template (not shown), the list option template being a template or sample for setting an interface or layout of a mini program. In an example, the user taps the mini program interface **200** to generate the next hierarchy of program interface **220**. In some embodiments, the user names the program interface **220** as a page 1. The operating option **111** is used for editing the mini program interface **220**, or performing a custom setting on the mini program interface **220**. For example, the user selects the operating option **111** to enter a mini program custom setting interface. The image changing option **112** is used for changing an image in a mini program. In some embodiments, an image is changed by calling a local gallery or album in a terminal by using a visualization production program platform, or an image in the mini program is changed by connecting to a network and obtaining an image from the network, or an image in the mini program is changed by receiving an image file.
(139) The user performs a data binding operation on the next hierarchy of program interface **220** by using the data binding option **107**, and the process is described with reference to FIG. **16**. FIG. **16** is a schematic interface diagram of performing data binding on a next hierarchy of program interface according to an exemplary embodiment of this application. A completed mini program interface **200**, a next hierarchy of program interface **220**, an image control component **221**, a binding page input parameter **222**, a delete option **108**, an operating option **111**, an image changing option **112**, and a preview control **129** are displayed on a data binding interface **19**.
(140) In some embodiments, the user taps the image control component **221**, and the data binding option **107** is displayed. The user taps the data binding option **107**, and the binding page input parameter **222** is displayed. In some embodiments, a none-binding option **2221** and an image data field option **2222** are displayed on the binding page input parameter **222**. In some embodiments, at least one of a title binding data field, a content binding data field, a video binding data field, and an animation binding data field is further displayed on the binding page input parameter **222**. In an example, the user selects the none-binding option **2221**, and in this case, the image control component **221** is not yet bound with any image data field. If the user selects the image data field option **2222**, the image control component **221** is bound with a data field corresponding to the image data field option **2222**, and an image is displayed at a position corresponding to the image control component **221**. In some embodiments, the image data field option **2222** further includes a table file name corresponding to the data field.
(141) In some embodiments, after the user completes the data binding operation on the next hierarchy of program interface **220**, the data field bound with the target control component is

automatically bound with a corresponding control component in the next hierarchy of program interface by using the visualization production program platform.

(142) For example, the user taps a list option on the mini program interface shown in FIG. **14**. For example, the user taps a first list option (corresponding to a column of "*The Adventures of Xiao Zhang*"), a next hierarchy of program interface (that is, a page 1 is jumped to from a homepage interface) is jumped to. After a target control component corresponding to the first list option is bound with a data field, the data field is automatically bound with a corresponding control component in the next hierarchy of program interface **220** (that is, the next hierarchy of program interface **220** displaying bound data, that is, a page displaying details).

(143) FIG. **17** is a schematic diagram of a preview interface of a next hierarchy of program interface according to an exemplary embodiment of this application. A completed mini program interface **200**, a next hierarchy of program interface **220**, an image control component **221**, preview prompt information **204**, and a preview control **129** are displayed on the preview interface **20**.

(144) In some embodiments, the user selects the preview control **129** to preview the next hierarchy of program interface **220**. The preview prompt information **204** is used for prompting the user that a mini program is pushed to a host application program. In some embodiments, the user may push the next hierarchy of program interface **220** to a host application program in a manual pushing manner, and the process is consistent with the process of pushing a mini program interface (that is, a homepage interface). This is not repeated herein.

(145) FIG. **18** is a schematic diagram of a next hierarchy of program interface on a terminal according to an exemplary embodiment of this application. An image **210**, a title **211**, and content **212** are displayed on the next hierarchy of program interface **220**. In some embodiments, the user taps the first list item (as shown in FIG. **14**, a title corresponding to the first list item is *The Adventures of Xiao Zhang*) on the homepage interface of the mini program, a page is jumped to the next hierarchy of program interface **220** (that is, a page 1, specific content about "*The Adventures of Xiao Zhang*" is displayed on the page 1; for example, this is a book worthy of recommendation, and describes the story of Xiao Zhang's journey).

(146) In a case that a data binding operation corresponding to a target basic UI control on the program interface of the mini program is received, and after the target basic UI control is bound with the data source according to the data binding operation, optionally, the user may select program code generating a mini program. The visualization production program platform generates view layer code of the mini program according to a basic UI control on the program interface of the mini program, for example, generates the view layer code of the mini program according to an image control component; generates logic layer code of the mini program according to a data source bound with a basic UI control on the program interface of the mini program, for example, generates logic layer code of the mini program according to an image data field bound with an image control component; and generates a program package of the mini program according to the view layer code and the logic layer code, then transmits the program package of the mini program to a host application program for at least one of a running operation and a preview operation, or transmits the program package of the mini program to a program platform of the host application program for auditing or publishing.

(147) For example, the visualization production program platform determines a list item template component for which code needs to be generated, recursively generates markup language (ML) code corresponding to the image control component, adds code (such as, loop statement code: for statement) representing a loop statement to the list item template component, and binds an attribute value with an array by using the code representing the loop statement, so as to repeatedly render the list item template component by using all data in the array, the attribute value being a name of a bound data table or a name corresponding to a bound data field. The visualization production program platform determines control components for which code needs to be generated, recursively generates each control component and corresponding ML code, checks whether the

control component is bound with a data field, and adds, in a case that the control component is bound with a data field, a name of the data field to the ML code. The list item template component is a template component corresponding to a structure or a frame that can hold data in a mini program interface. The recursive generation means that a program directly or indirectly calls an own program during running. A complex problem can be converted into a simple problem similar to the original problem to resolve only by using a small amount of program code.
(148) For example, the visualization production program platform converts a format of a bound data table into a JavaScript Object Notation (JSON) format, the format storing and representing data by using a text format independent of a programing language, which is convenient for a user to read and write. A computer can generate a file of the format or parse a file of the format more easily. The visualization production program platform places the data table converted into a JSON format into a data object corresponding to page logic code. When the program package of the mini program is pushed to the host application program for running, the user may see that the list item template component bound with the data field is rendered as an actual list (that is, a page of the mini program) together. Rendering means that a list item or a basic UI control and corresponding data are converted into content displayed in a corresponding program by using code.
(149) For example, when creating a relationship of jumping from a list item to a new page, the visualization production program platform adds, in a case that the list item is detected to be bound with a data field, the bound data field to code representing a jumped-to page, reads an index of the data field from an event response function corresponding to code (such as bindtap code) of a binding interaction type, obtains a data field corresponding to an actual obtained new page according to the index, and refreshes the data field to the new page by calling a set data (setData function refers to transmitting data from a logic layer to a view layer) method.
(150) A developer with professional knowledge may select to generate program code of a mini program, and modify content of the mini program by modifying corresponding code.
(151) FIG. **19** is a schematic interface diagram of logic editing according to an exemplary embodiment of this application. As shown in FIG. **19**(*a*), a code display region **300** and an image control component **103** are displayed on a logic editing interface **21**. A user operates according to a related code editing method when performing code editing on the code display region **300** and jumping from a current page to a next page, or viewing an attribute of a target UI control, and the user needs to edit code corresponding to all functions that the target UI control may implement. If the user does not have professional knowledge in the code editing, difficulty in the code editing is caused.
(152) In an example, in a case that a triggering operation of a user on a target control component is received, the visualization production program obtains a function option (for example, an attribute option **106**, a data binding option **107**, or a delete option **108**) corresponding to the target control component. Function code corresponding to the target control component is automatically typed into the code display region **300**, that is, the function code (such as function code) related to the currently edited target control component is automatically typed into a line of code requiring typing function code, thereby providing convenience for users having a code editing ability.
(153) In some embodiments, before receiving the triggering operation of the user on the target control component, the method further includes receiving a triggering operation generated in a code region in a case that function code corresponding to a target control component needs to be typed into. In some embodiments, the triggering operation on the target control component includes a tap operation, a double-tap operation, and the like on the target control component, and this is not limited in this embodiment of this application.
(154) For example, as shown in FIG. **19**(*b*), in the code display region **300**, content needs to be typed into a variation region **301** is function code of a current target control component. After a tap operation of the user on the variation region **301** is received, a tap operation of the user on the target control component is waited to be received. After a tap operation of the user on the image

control component **103** is received, the visualization production program displays function options corresponding to the image control component, for example, an attribute option **106**, a data binding option **107**, a delete option **108**, an operating option **111**, and an image changing option **112**. In this case, function code corresponding to the image control component **103** is automatically typed into and displayed on the variation region **301**, and therefore, the user does not need to manually type the function code of the image control component **103** in the variation region **301**.

(155) In some embodiments, after a mini program is produced and content of the mini program is confirmed to have no error, the user may transmit the mini program to a host application program. For example, the user taps a submitting control, and in a case that a submitting operation on the submitting control is received by a mini program production interface of the visualization production program, a program package of the mini program is transmitted to a program platform of the host application program for auditing or publishing. In some embodiments, the submitting control may be alternatively named as a submitting option or an uploading option, and this is not limited in this application. In some embodiments, after succeeding in auditing, the mini program may be used by other users.

(156) FIG. **20** is a structural block diagram of a terminal according to an exemplary embodiment of this application. A host application program **162** and a mini program **164** are installed and run on a user terminal. The host application program **162** may be any one of an instant messaging program, a social program, and a payment program. An example in which the terminal is an instant messaging program is used for description in this application. A mini program is a program executed in the host application program **162**. There may be a plurality of mini programs, one of which is a mini program produced by a user. The user enters the produced mini program **164** by using the host application program **162**. In some embodiments, the mini program produced by the user needs to obtain information (for example, an account, a communication number, and a geographic location of the user) of the user or does not need to obtain the information of the user.

(157) The host application program **162** is an application for a user to produce a mini program **164**, and provides an environment for the user to produce the mini program **164**. The host application program **162** is a native application. The native application is an application that may be directly run on an operating system **161**. The host application program **162** may be a social application, a dedicated application specially supporting the production of the mini program **164**, a file management application, an email application, a game application, or the like. The social application includes an instant messaging application (for example, WeChat and QQ), a social network service (SNS) application, a live broadcast application, or the like. The mini program is an application that may be run in an environment provided by the host application program. The mini program may be a social application, a file management application, a mail application, a game application, or the like. An example in which the mini program is the mini program **164** produced by the user is used for description. The host application program **162** may be a WeChat program.

(158) A mini program logic layer unit **164***b* and a corresponding mini program view layer unit **164***a* are configured to implement an instance of the mini program. One mini program may be implemented by using one mini program logic layer unit **164***b* or at least one mini program view layer unit **164***a*. The mini program view layer unit **164***a* and a mini program page may be in a one-to-one correspondence.

(159) The mini program view layer unit **164***a* is configured to organize and render a view of the mini program. The mini program logic layer unit **164***b* is configured to process a data processing logic of the mini program and the corresponding mini program page. A unit may be a process or a thread. For example, the mini program view layer unit **164***a* is a mini program view layer thread, and the mini program logic layer unit **164***b* is a mini program logic layer thread. The mini program logic layer unit **164***b* unit may be run in a virtual machine. The mini program view layer unit **164***a* and the mini program logic layer unit **164***b* may perform relay communication with each other by using a host application program native unit **162***a*. The host application program native unit **162***a* is

an interface for the host application program **162** and the mini program to communicate with each other, and the host application program native unit **162***a* may be a thread or process of the host application program **162**. Page logic code of a program package to which each mini program page pertains may be registered when the mini program logic layer unit **164***b* is started, and the registered page logic code is executed when the page logic code is needed to process data.
(160) A method, system, and service scenario described in the embodiments of this application is for more clearly describing the technical solutions in the embodiments of this application, and does not constitute a limitation on the technical solutions provided in the embodiments of this application. Persons of ordinary skill may learn that, with evolution of technologies and appearance of a new service scenario, the technical solutions provided in the embodiments of this application also apply to a similar technical problem.
(161) The following is an apparatus embodiment of this application. For details not described in detail in the apparatus embodiment, reference may be made to corresponding record in the foregoing method embodiments. Details are not repeated herein.
(162) FIG. **21** is a schematic structural diagram of a mini program data binding apparatus according to an exemplary embodiment of this application. The apparatus may be implemented as the terminal or a part of the terminal by means of software, hardware, or a combination thereof. The apparatus includes: a display module **2110**, a receiving module **2120**, an obtaining module **2130**, and a control module **2140**. The receiving module **2120** is an optional module.
(163) The display module **2110** is configured to display a mini program production interface of a visualization production program including a panel region and an editing region, the panel region being provided with n types of basic UI controls, n being a positive integer.
(164) The receiving module **2120** is configured to generate, in response to receiving a user operation on a selected basic UI control, a program interface of the mini program in the editing region according to the selected basic UI control.
(165) The obtaining module **2130** is configured to obtain a data source.
(166) The receiving module **2120** is further configured to bind, in response to receiving a data binding operation corresponding to a target basic UI control on the program interface of the mini program, the target basic UI control with the data source, the data source being used for displaying the target basic UI control.
(167) In one implementation, the basic UI control has m corresponding control components, and the data source includes m data fields, m being a positive integer greater than 1.
(168) The display module **2110** is configured to display, in response to receiving the data binding operation, binding preview information that binds the m data fields with the m control components respectively according to a first arrangement order.
(169) The display module **2110** is configured to display, in response to receiving a rearrangement operation, binding preview information that binds the m data fields with the m control components respectively according to an (i+1).sup.th arrangement order, the (i+1).sup.th arrangement order being an arrangement order different from an i.sup.th arrangement order, i being a positive integer with an initial value being 1, and the i.sup.th arrangement order being generated randomly or preset in advance.
(170) The receiving module **2120** is configured to bind, in response to receiving a binding confirmation operation, the m data fields with the m control components respectively.
(171) In one implementation, the control module **2140** is configured to generate all arrangement orders of the m data fields; the receiving module **2120** is configured to determine, in response to receiving the rearrangement operation, an arrangement order, different from the i.sup.th arrangement order, from all the arrangement orders randomly as the (i+1).sup.th arrangement order; and the display module **2110** is configured to display the binding preview information that binds the m data fields with the m control components respectively according to the (i+1).sup.th arrangement order.

(172) In one implementation, the obtaining module **2130** is configured to read a j.sup.th data field and a field type thereof from the m data fields according to the (i+1).sup.th arrangement order, 1≤j≤m, and j being a positive integer; the control module **2140** is configured to determine a target control component satisfying a binding condition from the m control components, the binding condition including that: the control component is not yet bound with a data field; or the control component matches the field type of the j.sup.th data field; or the control component is not yet bound with a data field and matches the field type of the j.sup.th data field; and bind the target control component with the j.sup.th data field; and the display module **2110** is configured to display, after the m control components are all bound, the binding preview information that binds the m data fields with the m control components respectively.
(173) In one implementation, the receiving module **2120** is configured to receive a binding modification operation on the target control component on the target basic UI control; the display module **2110** is configured to display at least one candidate data field corresponding to the target control component, the candidate data field being a data field, whose field type matches the target control component, in the m data fields; and the control module **2140** is configured to bind, in response to receiving a selection operation, a data field selected from the at least one candidate data field with the target control component.
(174) In one implementation, the mini program further includes a next hierarchy of program interface, the next hierarchy of program interface being a program interface displayed after the target basic UI control is triggered; and the control module **2140** is configured to bind the data field bound with the target control component with a corresponding control component in the next hierarchy of program interface automatically.
(175) In one implementation, the display module **2110** is configured to display a first table file; and the receiving module **2120** is configured to receive an addition operation, the addition operation being used for adding the first table file to the visualization production program; or the receiving module **2120** is configured to receive a program calling operation, the program calling operation being used for calling a table application in the visualization production program, and the table application including a second table file; or the receiving module **2120** is configured to receive a third table file by using a short range wireless communication technology.
(176) In one implementation, the control module **2140** is configured to generate view layer code of the mini program according to the basic UI control on the program interface of the mini program; generate logic layer code of the mini program according to the data source bound with the basic UI control on the program interface of the mini program; and generate a program package of the mini program according to the view layer code and the logic layer code.
(177) In one implementation, the mini program production interface further includes a preview control; and the receiving module **2120** is configured to push, in response to receiving a preview operation on the preview control, the program package of the mini program to the host application program for running and/or previewing.
(178) In one implementation, the mini program production interface further includes a submitting control; and the receiving module **2120** is configured to transmit, in response to receiving a submitting operation on the submitting control, the program package of the mini program to a program platform of the host application program for auditing or publishing.
(179) In conclusion, according to the mini program data binding apparatus provided in this embodiment, an operation is performed on a mini program production interface of a visualization production program, a target basic UI control of the mini program is bound with a data source, and the target basic UI control undergone binding can display content corresponding to the data source. Therefore, a user may implement a function of editing or modifying displayed content in the target basic UI control by changing the data source bound with the target basic UI control, or unbinding a binding relationship between the target basic UI control and the data source, without writing code, thereby improving efficiency of a user for producing a mini program. In this application, the term

“unit” or “module” refers to a computer program or part of the computer program that has a predefined function and works together with other related parts to achieve a predefined goal and may be all or partially implemented by using software, hardware (e.g., processing circuitry and/or memory configured to perform the predefined functions), or a combination thereof. Each unit or module can be implemented using one or more processors (or processors and memory). Likewise, a processor (or processors and memory) can be used to implement one or more modules or units. Moreover, each module or unit can be part of an overall module that includes the functionalities of the module or unit.

(180) A computer device used in this application is described in the following. FIG. **22** is a structural block diagram of a computer device **2200** according to an exemplary embodiment of this application. The computer device **2200** may be a portable mobile terminal, for example, a smartphone, a tablet computer, a moving picture experts group audio layer III (MP3) player, or a moving picture experts group audio layer IV (MP4) player. The computer device **2200** may be further referred to as other names such as user equipment and a portable terminal.

(181) Generally, the computer device **2200** includes a processor **2201** and a memory **2202**.

(182) The processor **2201** may include one or more processing cores. For example, the processor may be a 4-core processor or an 8-core processor. The processor **2201** may be implemented in at least one hardware form of digital signal processor (DSP), a field-programmable gate array (FPGA), and a programmable logic array (PLA). The processor **2201** may alternatively include a main processor and a coprocessor. The main processor is configured to process data in an active state, also referred to as a central processing unit (CPU). The coprocessor is a low-power processor configured to process data in a standby state. In some embodiments, the processor **2201** may be integrated with a graphics processing unit (GPU). The GPU is configured to render and draw content that needs to be displayed on a display screen. In some embodiments, the processor **2201** may further include an artificial intelligence (AI) processor. The AI processor is configured to process a computing operation related to machine learning.

(183) The memory **2202** may include one or more computer-readable storage media. The computer-readable storage medium may be tangible and non-transient. The memory **2202** may further include a high-speed random access memory (RAM), and a non-volatile memory such as one or more magnetic disk storage devices and a flash storage device. In some embodiments, a non-transitory computer-readable storage medium in the memory **2202** is configured to store at least one instruction, the at least one instruction being configured to be executed by the processor **2201** to implement the mini program data binding method provided in this application.

(184) In some embodiments, the computer device **2200** may include: a peripheral interface **2203** and at least one peripheral. Specifically, the peripheral includes: at least one of a radio frequency (RF) circuit **2204**, a touch display screen **2205**, a camera component **2206**, an audio circuit **2207**, a positioning component **2208**, and a power supply **2209**.

(185) The peripheral interface **2203** may be configured to connect the at least one peripheral related to input/output (I/O) to the processor **2201** and the memory **2202**. In some embodiments, the processor **2201**, the memory **2202**, and the peripheral interface **2203** are integrated on the same chip or the same circuit board. In some other embodiments, any or both of the processor **2201**, the memory **2202**, and the peripheral interface **2203** may be implemented on an independent chip or circuit board. This is not limited in this embodiment.

(186) The RF circuit **2204** is configured to receive and transmit an RF signal, which is also referred to as an electromagnetic signal. The RF circuit **2204** communicates with a communication network and another communication device by using the electromagnetic signal. The RF circuit **2204** converts an electric signal into an electromagnetic signal for transmission, or converts a received electromagnetic signal into an electric signal. In some embodiments, the RF circuit **2204** includes: an antenna system, an RF transceiver, one or more amplifiers, a tuner, an oscillator, a digital signal processor, a codec chip set, a subscriber identity module card, and the like. The RF circuit **2204**

may communicate with another terminal by using at least one wireless communication protocol. The wireless communication protocol includes, but is not limited to: a world wide web, a metropolitan area network, an intranet, generations of mobile communication networks (2G, 3G, 4G, and 5G), a wireless local area network, and/or a Wi-Fi network. In some embodiments, the RF circuit **2204** may further include a circuit related to near field communication (NFC), which is not limited in this application.
(187) The touch display screen **2205** is configured to display a UI. The UI may include a graph, text, an icon, a video, and any combination thereof. The touch display screen **2205** also has a capability of collecting a touch signal on or above a surface of the touch display screen **2205**. The touch signal may be used as a control signal to be inputted to the processor **2201** for processing. The touch display screen **2205** is configured to provide a virtual button and/or a virtual keyboard, which is also referred to as a soft button and/or a soft keyboard. In some embodiments, there may be one touch display screen **2205**, disposed on a front panel of the computer device **2200**. In some other embodiments, there may be at least two touch display screens **2205**, disposed on different surfaces of the computer device **2200** respectively or in a folded design. In some embodiments, the touch display screen **2205** may be a flexible display screen, disposed on a curved surface or a folded surface of the computer device **2200**. Even, the touch display screen **2205** may be further set in a non-rectangular irregular pattern, namely, a special-shaped screen. The touch display screen **2205** may be made of a material such as a liquid crystal display (LCD) or an organic light-emitting diode (OLED).
(188) The camera component **2206** is configured to capture images or videos. In some embodiments, the camera component **2206** includes a front-facing camera and a rear-facing camera. Generally, the front-facing camera is configured to implement a video call or self-portrait. The rear-facing camera is configured to capture a picture or a video. In some embodiments, there are at least two rear-facing cameras, each of which is any one of a main camera, a depth of field camera and a wide-angle camera, to implement a background blurring function by fusing the main camera and the depth of field camera, and panoramic shooting and virtual reality (VR) shooting functions by fusing the main camera and the wide-angle camera. In some embodiments, the camera component **2206** may further include a flashlight. The flashlight may be a monochrome temperature flashlight, or may be a double color temperature flashlight. The double color temperature flash refers to a combination of a warm light flash and a cold light flash, and may be used for light compensation under different color temperatures.
(189) The audio circuit **2207** is configured to provide an audio interface between a user and the computer device **2200**. The audio circuit **2207** may include a microphone and a speaker. The microphone is configured to collect sound waves of users and surroundings, and convert the sound waves into electrical signals and input the signals to the processor **2201** for processing, or input the signals into the RF circuit **2204** to implement voice communication. For the purpose of stereo sound collection or noise reduction, there may be a plurality of microphones, respectively disposed at different parts of the computer device **2200**. The microphone may be further a microphone array or an omnidirectional acquisition microphone. The speaker is configured to convert electric signals from the processor **2201** or the RF circuit **2204** into sound waves. The speaker may be a conventional thin-film speaker or a piezoelectric ceramic speaker. When the speaker is the piezoelectric ceramic speaker, electric signals not only may be converted into sound waves that can be heard by human, but also may be converted into sound waves that cannot be heard by human for ranging and the like. In some embodiments, the audio circuit **2207** may also include an earphone jack.
(190) The positioning component **2208** is configured to determine a current geographic location of the computer device **2200** through positioning, to implement navigation or a location based service (LBS). The positioning component **2208** may be a positioning assembly based on the Global Positioning System (GPS) of the United States, the China's Beidou Navigation Satellite System

(BDS), the GLONASS System of Russia, or the GALILEO System of the European Union.
(191) The power supply **2209** is configured to supply power to components in the computer device **2200**. The power supply **2209** may be an alternating-current power supply, a direct-current power supply, a disposable battery, or a rechargeable battery. When the power supply **2209** includes the rechargeable battery, the rechargeable battery may be a wired charging battery or a wireless charging battery. The wired charging battery is a battery charged through a wired line, and the wireless charging battery is a battery charged through a wireless coil. The rechargeable battery may be further configured to support a quick charge technology.
(192) In some embodiments, the computer device **2200** may also include one or more sensors **2210**. The one or more sensors **2210** include, but are not limited to, an acceleration sensor **2211**, a gyroscope sensor **2212**, a pressure sensor **2213**, a fingerprint sensor **2214**, an optical sensor **2215**, and a proximity sensor **2216**.
(193) The acceleration sensor **2211** may detect accelerations on three coordinate axes of a coordinate system established by the computer device **2200**. For example, the acceleration sensor **2211** may be configured to detect components of gravity acceleration on the three coordinate axes. The processor **2201** may control, according to a gravity acceleration signal collected by the acceleration sensor **2211**, the touch display screen **2205** to display the UI in a landscape view or a portrait view. The acceleration sensor **2211** may be further configured to collect motion data of a game or a user.
(194) The gyroscope sensor **2212** may detect a body direction and a rotation angle of the computer device **2200**. The gyroscope sensor **2212** may cooperate with the acceleration sensor **2211** to collect a 3D action by the user on the computer device **2200**. The processor **2201** may implement the following functions according to the data collected by the gyroscope sensor **2212**: motion sensing (such as changing the UI according to a tilt operation of the user), image stabilization during shooting, game control, and inertial navigation.
(195) The pressure sensor **2213** may be disposed on a side frame of the computer device **2200** and/or a lower layer of the touch display screen **2205**. When the pressure sensor **2213** is disposed at the side frame of the computer device **2200**, a holding signal of the user on the computer device **2200** may be detected, and left/right hand identification and a quick action may be performed according to the holding signal. When the pressure sensor **2213** is disposed at the lower layer of the touch display screen **2205**, an operable control on the UI interface can be controlled according to a pressure operation of the user on the touch display screen **2205**. The operable control includes at least one of a button control, a scroll-bar control, an icon control, and a menu control.
(196) The fingerprint sensor **2214** is configured to acquire a user's fingerprint to identify a user's identity according to the acquired fingerprint. When identifying that the identity of the user is a trusted identity, the processor **2201** authorizes the user to perform related sensitive operations. The sensitive operations include: unlocking a screen, viewing encryption information, downloading software, paying and changing a setting, and the like. The fingerprint sensor **2214** may be disposed on a front face, a back face, or a side face of the computer device **2200**. When a physical button or a vendor logo is disposed on the computer device **2200**, the fingerprint sensor **2214** may be integrated together with the physical button or the vendor logo.
(197) The optical sensor **2215** is configured to collect ambient light intensity. In an embodiment, the processor **2201** may control display luminance of the touch display screen **2205** according to the ambient light intensity collected by the optical sensor **2215**. Specifically, in a case that the ambient light intensity is relatively high, the display luminance of the touch display screen **2205** is increased. In a case that the ambient light intensity is relatively low, the display luminance of the touch display screen **2205** is reduced. In another embodiment, the processor **2201** may further dynamically adjust a camera parameter of the camera component **2206** according to the ambient light intensity collected by the optical sensor **2215**.
(198) The proximity sensor **2216**, also referred to as a distance sensor, is usually disposed on the

front panel of the computer device **2200**. The proximity sensor **2216** is configured to collect a distance between a front face of the user and the front face of the computer device **2200**. In an embodiment, when the proximity sensor **2216** detects that the distance between the user and the front surface of the computer device **2200** gradually becomes small, the touch display screen **2205** is controlled by the processor **2201** to switch from a screen-on state to a screen-off state. When the proximity sensor **2216** detects that the distance between the user and the front surface of the computer device **2200** gradually becomes large, the touch display screen **2205** is controlled by the processor **2201** to switch from the screen-off state to the screen-on state.
(199) A person skilled in the art may understand that the structure shown in FIG. **21** does not constitute any limitation on the computer device **2100**, and the computer device may include more components or fewer components than those shown in the figure, or some components may be combined, or a different component deployment may be used.
(200) This application further provides a computer device including a processor and a memory, the memory storing at least one instruction, at least one segment of program, a code set, or an instruction set, the at least one instruction, the at least one segment of program, the code set, or the instruction set being loaded and executed by the processor to implement the mini program data binding method provided in the foregoing method embodiments.
(201) This application further provides a computer-readable storage medium, storing at least one instruction, at least one program, a code set, or an instruction set, the at least one instruction, the at least one program, the code set, or the instruction set being loaded and executed by a processor to implement the mini program data binding method provided in the foregoing method embodiments.
(202) This application further provides a computer program product or a computer program. The computer program product or the computer program includes computer instructions, and the computer instructions are stored in a computer-readable storage medium. A processor of a computer device reads the computer instructions from the computer-readable storage medium, and executes the computer instructions, so that the computer device performs the mini program data binding method provided in optional implementations of the foregoing aspect.
(203) “Plurality of” mentioned in this specification means two or more. “And/or” describes an association relationship for associated objects and represents that three relationships may exist. For example, A and/or B may represent the following three cases: only A exists, both A and B exist, and only B exists. The character “/” in this specification generally indicates an “or” relationship between the associated objects.
(204) A person of ordinary skill in the art may understand that all or some of the steps of the foregoing embodiments may be implemented by hardware, or may be implemented a program instructing related hardware. The program may be stored in a computer-readable storage medium. The storage medium may be: a ROM, a magnetic disk, or an optical disc.
(205) The foregoing descriptions are merely illustrative embodiments of this application, but are not intended to limit this application. Any modification, equivalent replacement, or improvement made within the spirit and principle of this application shall fall within the protection scope of this application.

Claims

1. A mini program data binding method performed at a terminal, the mini program being a program executed in a host application program, and the method comprising: displaying a mini program production interface of a visualization production program including a panel region and an editing region, the panel region being provided with n types of basic user interface (UI) controls, n being a positive integer, and each basic UI control having m corresponding control components, m being a positive integer greater than 1, each control component having an associated data type; in response to receiving a user operation on a selected basic UI control, generating a program interface of the

mini program in the editing region according to the user-selected basic UI control; obtaining a data source, wherein the data source comprises a plurality of data entries, each data entry having the m data fields; displaying metadata of the data source in the mini program production interface, the metadata including each of the m data fields and its corresponding data type; in response to receiving a data binding operation corresponding to a target basic UI control on the program interface of the mini program, randomly binding a respective one of the control components of the target basic UI control with a corresponding one of the m data fields of the data source having the data type, the data source being used for displaying the target basic UI control; and updating the program interface with one of the plurality of data entries, wherein each control component of the target basic UI control is populated with a value of a corresponding data field bound to the control component; in response to receiving a preview operation, generating a view layer code of the mini program according to the user-selected basic UI control on the program interface of the mini program; generating a logic layer code of the mini program according to the data source bound with the selected basic UI control on the program interface of the mini program; generating a program package of the mini program according to the view layer code and the logic layer code; and transmitting the program package of the mini program to the host application program for running and/or previewing.

2. The method according to claim 1, wherein the randomly binding a respective one of the control components of the target basic UI control with a corresponding one of the m data fields of the data source, the data source being used for displaying the target basic UI control comprises: displaying, in response to receiving the data binding operation, binding preview information that binds each of the m data fields with a corresponding one of the m control components respectively according to a first random arrangement order; displaying, in response to receiving a rearrangement operation, binding preview information that binds each of the m data fields with another corresponding one of the m control components respectively according to an (i+1).sup.th random arrangement order, the (i+1).sup.th arrangement order being an arrangement order different from an i.sup.th arrangement order, i being a positive integer with an initial value being 1; and binding, in response to receiving a binding confirmation operation, the m data fields with the m control components respectively.

3. The method according to claim 2, wherein the method further comprises: generating all arrangement orders of the m data fields; and the displaying, in response to receiving a rearrangement operation, binding preview information that binds each of the m data fields with another corresponding one of the m control components respectively according to an (i+1).sup.th arrangement order comprises: determining, in response to receiving the rearrangement operation, an arrangement order, different from the i.sup.th arrangement order, from all the arrangement orders randomly as the (i+1).sup.th arrangement order; and displaying the binding preview information that binds the m data fields with the m control components respectively according to the (i+1).sup.th arrangement order.

4. The method according to claim 3, wherein the displaying the binding preview information that binds the m data fields with the m control components respectively according to the (i+1).sup.th arrangement order comprises: reading a j.sup.th data field and a field type thereof from the m data fields according to the (i+1).sup.th arrangement order, 1≤j≤m, and j being a positive integer; determining a target control component satisfying a binding condition from the m control components, the binding condition comprising that: the control component is not yet bound with a data field; the control component matches the field type of the j.sup.th data field; or the control component is not yet bound with a data field and matches the field type of the j.sup.th data field; binding the target control component with the j.sup.th data field; and displaying, after the m control components are all bound, the binding preview information that binds the m data fields with the m control components respectively.

5. The method according to claim 2, wherein the method further comprises: receiving a binding modification operation on the target control component on the target basic UI control; displaying at

least one candidate data field corresponding to the target control component, the candidate data field being a data field, whose field type matches the target control component, in the m data fields; and binding, in response to receiving a selection operation, a data field selected from the at least one candidate data field with the target control component.

6. The method according to claim 2, wherein the mini program further comprises a next hierarchy of program interface, the next hierarchy of program interface being a program interface displayed after the target basic UI control is triggered, and the method further comprises: binding the data field bound with the target control component with a corresponding control component in the next hierarchy of program interface automatically.

7. The method according to claim 1, wherein the obtaining a data source comprises: displaying a first table file; and receiving an addition operation, the addition operation being used for adding the first table file to the visualization production program; or receiving a program calling operation, the program calling operation being used for calling a table application in the visualization production program, the table application comprising a second table file; or receiving a third table file by using a short range wireless communication technology.

8. The method according to claim 1, wherein the mini program production interface further comprises a submitting control; and the method further comprises: transmitting, in response to receiving a submitting operation on the submitting control, the program package of the mini program to a program platform of the host application program for auditing or publishing.

9. A computer device, comprising a processor and a memory, the memory storing at least one instruction, the at least one instruction being loaded and executed by the processor to perform a plurality of operations for binding mini program data, the plurality of operations including: displaying a mini program production interface of a visualization production program including a panel region and an editing region, the panel region being provided with n types of basic user interface (UI) controls, n being a positive integer, and each basic UI control having m corresponding control components, m being a positive integer greater than 1, each control component having an associated data type; in response to receiving a user operation on a selected basic UI control, generating a program interface of the mini program in the editing region according to the user-selected basic UI control; obtaining a data source, wherein the data source comprises a plurality of data entries, each data entry having the m data fields; displaying metadata of the data source in the mini program production interface, the metadata including each of the m data fields and its corresponding data type; in response to receiving a data binding operation corresponding to a target basic UI control on the program interface of the mini program, randomly binding a respective one of the control components of the target basic UI control with a corresponding one of the m data fields of the data source having the data type, the data source being used for displaying the target basic UI control; and updating the program interface with one of the plurality of data entries, wherein each control component of the target basic UI control is populated with a value of a corresponding data field bound to the control component; in response to receiving a preview operation, generating a view layer code of the mini program according to the user-selected basic UI control on the program interface of the mini program; generating a logic layer code of the mini program according to the data source bound with the selected basic UI control on the program interface of the mini program; generating a program package of the mini program according to the view layer code and the logic layer code; and transmitting the program package of the mini program to the host application program for running and/or previewing.

10. The computer device according to claim 9, wherein the randomly binding a respective one of the control components of the target basic UI control with a corresponding one of the m data fields of the data source, the data source being used for displaying the target basic UI control comprises: displaying, in response to receiving the data binding operation, binding preview information that binds each of the m data fields with a corresponding one of the m control components respectively according to a first random arrangement order; displaying, in response to receiving a rearrangement

operation, binding preview information that binds each of the m data fields with another corresponding one of the m control components respectively according to an (i+1).sup.th random arrangement order, the (i+1).sup.th arrangement order being an arrangement order different from an i.sup.th arrangement order, i being a positive integer with an initial value being 1; and binding, in response to receiving a binding confirmation operation, the m data fields with the m control components respectively.

11. The computer device according to claim 10, wherein the plurality of operations further comprise: generating all arrangement orders of the m data fields; and the displaying, in response to receiving a rearrangement operation, binding preview information that binds each of the m data fields with another corresponding one of the m control components respectively according to an (i+1).sup.th arrangement order comprises: determining, in response to receiving the rearrangement operation, an arrangement order, different from the i.sup.th arrangement order, from all the arrangement orders randomly as the (i+1).sup.th arrangement order; and displaying the binding preview information that binds the m data fields with the m control components respectively according to the (i+1).sup.th arrangement order.

12. The computer device according to claim 11, wherein the displaying the binding preview information that binds the m data fields with the m control components respectively according to the (i+1).sup.th arrangement order comprises: reading a j.sup.th data field and a field type thereof from the m data fields according to the (i+1).sup.th arrangement order, $1 \leq j \leq m$, and j being a positive integer; determining a target control component satisfying a binding condition from the m control components, the binding condition comprising that: the control component is not yet bound with a data field; the control component matches the field type of the j.sup.th data field; or the control component is not yet bound with a data field and matches the field type of the j.sup.th data field; binding the target control component with the j.sup.th data field; and displaying, after the m control components are all bound, the binding preview information that binds the m data fields with the m control components respectively.

13. The computer device according to claim 10, wherein the plurality of operations further comprise: receiving a binding modification operation on the target control component on the target basic UI control; displaying at least one candidate data field corresponding to the target control component, the candidate data field being a data field, whose field type matches the target control component, in the m data fields; and binding, in response to receiving a selection operation, a data field selected from the at least one candidate data field with the target control component.

14. The computer device according to claim 10, wherein the mini program further comprises a next hierarchy of program interface, the next hierarchy of program interface being a program interface displayed after the target basic UI control is triggered, and the plurality of operations further comprise: binding the data field bound with the target control component with a corresponding control component in the next hierarchy of program interface automatically.

15. The computer device according to claim 9, wherein the obtaining a data source comprises: displaying a first table file; and receiving an addition operation, the addition operation being used for adding the first table file to the visualization production program; or receiving a program calling operation, the program calling operation being used for calling a table application in the visualization production program, the table application comprising a second table file; or receiving a third table file by using a short range wireless communication technology.

16. The computer device according to claim 9, wherein the mini program production interface further comprises a submitting control; and the method further comprises: transmitting, in response to receiving a submitting operation on the submitting control, the program package of the mini program to a program platform of the host application program for auditing or publishing.

17. A non-transitory computer-readable storage medium, storing at least one instruction, the at least one instruction being loaded and executed by a processor of a computer device to perform a plurality of operations for binding mini program data, the plurality of operations including:

displaying a mini program production interface of a visualization production program including a panel region and an editing region, the panel region being provided with n types of basic user interface (UI) controls, n being a positive integer, and each basic UI control having m corresponding control components, m being a positive integer greater than 1, each control component having an associated data type; in response to receiving a user operation on a selected basic UI control, generating a program interface of the mini program in the editing region according to the user-selected basic UI control; obtaining a data source, wherein the data source comprises a plurality of data entries, each data entry having the m data fields; displaying metadata of the data source in the mini program production interface, the metadata including each of the m data fields and its corresponding data type; in response to receiving a data binding operation corresponding to a target basic UI control on the program interface of the mini program, randomly binding a respective one of the control components of the target basic UI control with a corresponding one of the m data fields of the data source having the data type, the data source being used for displaying the target basic UI control; and updating the program interface with one of the plurality of data entries, wherein each control component of the target basic UI control is populated with a value of a corresponding data field bound to the control component; in response to receiving a preview operation, generating a view layer code of the mini program according to the user-selected basic UI control on the program interface of the mini program; generating a logic layer code of the mini program according to the data source bound with the selected basic UI control on the program interface of the mini program; generating a program package of the mini program according to the view layer code and the logic layer code; and transmitting the program package of the mini program to the host application program for running and/or previewing.
18. The non-transitory computer-readable storage medium according to claim 17, wherein the randomly binding a respective one of the control components of the target basic UI control with a corresponding one of the m data fields of the data source, the data source being used for displaying the target basic UI control comprises: displaying, in response to receiving the data binding operation, binding preview information that binds each of the m data fields with a corresponding one of the m control components respectively according to a first random arrangement order; displaying, in response to receiving a rearrangement operation, binding preview information that binds each of the m data fields with another corresponding one of the m control components respectively according to an (i+1).sup.th random arrangement order, the (i+1).sup.th arrangement order being an arrangement order different from an i.sup.th arrangement order, i being a positive integer with an initial value being 1; and binding, in response to receiving a binding confirmation operation, the m data fields with the m control components respectively.
19. The non-transitory computer-readable storage medium according to claim 17, wherein the mini program production interface further comprises a submitting control; and the method further comprises: transmitting, in response to receiving a submitting operation on the submitting control, the program package of the mini program to a program platform of the host application program for auditing or publishing.